

Example
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Handwritten
Solved

$$y^4(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 24$$

The General form of Higher order differential equation is

$$y^{(n)}(t) + a_1 y^{(n-1)}(t) + a_2 y^{(n-2)}(t) + a_3 y^{(n-3)}(t) + \dots + a_n y(t) = b$$

$$y_p = b/a_n$$

Here $a_1 = 6, a_2 = 14, a_3 = 16, a_4 = 8$

$$y_p = b/a_4 = 24/8 = 3$$

$b = 24$

In order to find the required characteristic roots are.
The Characteristic equation is

$$r^4 + 6r^3 + 14r^2 + 16r + 8 = 0$$

Putting $r = -1$

$$(-1)^4 + 6(-1)^3 + 14(-1)^2 + 16(-1) + 8 = 0$$

$$1 - 6 + 14 - 16 + 8 = 0$$

$$23 - 22 = 1 \neq 0$$

Putting $r = -2$

$$(-2)^4 + 6(-2)^3 + 14(-2)^2 + 16(-2) + 8 = 0$$

$$16 - 48 + 56 - 32 + 8 = 0$$

$$80 - 80 = 0$$

Thus $r = -2$ is the root where $r = -2$ is the factor in r -terms. Here $(r+2)$ is the factor we find other factors

$$r+2 \sqrt{x^4 + 6x^3 + 14x^2 + 16x + 8} \quad \left[\begin{array}{l} x^3 + 4x^2 + 6x + 4 \\ + x^4 + 2x^3 \end{array} \right]$$

$$\begin{array}{r} 4x^3 + 14x^2 \\ + 4x^3 + 8x^2 \end{array}$$

$$\begin{array}{r} 6x^2 + 16x \\ + 6x^2 + 12x \end{array}$$

$$\begin{array}{r} 4x + 8 \\ + 4x + 8 \\ \hline 0 \end{array}$$

$$\begin{array}{r} r+2 \\ r^3 \\ \hline r^4 + 2r^2 \end{array}$$

$$(r+2)(x^3 + 4x^2 + 6x + 4) = 0$$

put $x = -1$

$$(-1)^3 + 4(-1)^2 + 6(-1) + 4 = 0$$

$$-1 + 4 - 6 + 4 = 0$$

$$8 - 7 = 1 \neq 0$$

put $x = -2$

$$(-2)^3 + 4(-2)^2 + 6(-2) + 4 = 0$$

$$-8 + 16 - 12 + 4 = 0$$

$$-20 + 20 = 0 \quad 0 = 0$$

$$r+2 \sqrt{x^3 + 4x^2 + 6x + 4} \quad \left[\begin{array}{l} x^2 + 2x + 2 \\ + x^2 + 2x^2 \end{array} \right]$$

$$\begin{array}{r} 2x^2 + 6x \\ + 2x^2 + 4x \end{array}$$

$$\begin{array}{r} 2x + 4 \\ 2x + 4 \\ \hline 0 \end{array}$$

The Character eq $x^4 + 6x^3 + 11x^2 + 6x + 8 = 0$
 can be factorized as

$$(x+2)(x+2)(x^2+2x+2)=0$$

(3)
 p.p.

$$x_1 = x_2 = -2 \text{ (Repeated)}$$

$$ax^2 + bx + c = 0$$

$$x^2 + 2x + 2$$

$$x_3, x_4 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a=1 \quad b=2 \quad c=2$$

$$\frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -\frac{2}{2} + \frac{2i}{2}$$

$$x_3, x_4 = -1 \pm i$$

$$h = -1 \quad v = 1$$

$$x_3, x_4 = h \pm vi$$

(Complex)

Thus

$$y_c = A_1 e^{x_1 t} + A_2 t e^{x_1 t} + A_3 e^{h t} (A_3 \cos t + A_4 \sin t)$$

$$y_c = A_1 e^{-2t} + A_2 t e^{-2t} + e^{-t} (A_3 \cos t + A_4 \sin t)$$

Hence the General Solution is

$$y(t) = y_c + y_p$$

$$y(t) = A_1 e^{-2t} + A_2 t e^{-2t} + e^{-t} (A_3 \cos t + A_4 \sin t) + 3$$